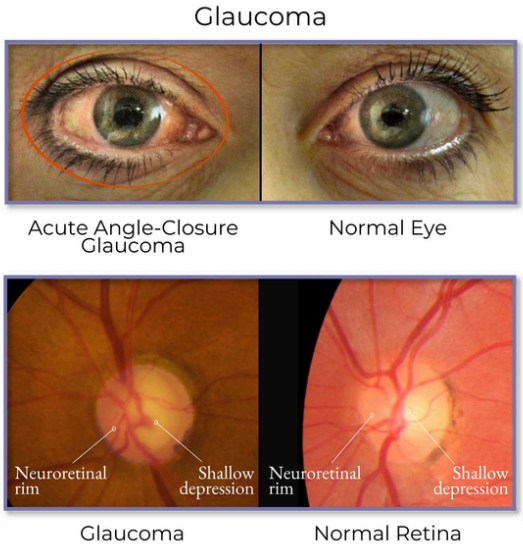
A COMPREHENSIVE DIGITAL IMAGE PROCESSING AND DEEP LEARNING FRAMEWORK FOR AUTOMATED GLAUCOMA DETECTION FROM FUNDUS IMAGES

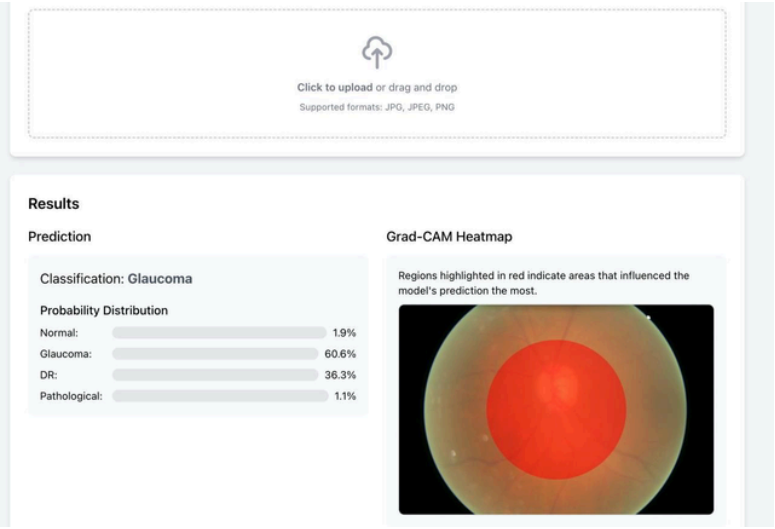
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Abstract

Glaucoma is a progressive optic neuropathy and the second leading cause of irreversible blindness worldwide, requiring early and reliable detection to prevent long-term vision loss. Traditional screening relies on manual evaluation of retinal fundus images, which is resource-intensive, subjective, and difficult to scale for population-level screening. This paper presents a comprehensive Digital Image Processing (DIP) and deep learning framework for automated glaucoma detection from fundus images. The proposed system integrates a multi-stage preprocessing pipeline—comprising optic disc–centered Region of Interest (ROI) extraction, Contrast Limited Adaptive Histogram Equalization (CLAHE), and Non-Local Means denoising—to enhance structural visibility and standardize image quality. A lightweight transfer learning classifier based on EfficientNet-B0 is fine-tuned to perform four-class glaucoma staging: Normal, Suspicious, Early, and Advanced. Additionally, Grad-CAM visualization is incorporated to provide clinically meaningful explainability by highlighting optic disc and cup regions that influence model predictions. Experimental results demonstrate that the combined DIP + deep learning pipeline yields significant improvements in accuracy, robustness, and interpretability, achieving strong performance across all stages and outperforming models trained on raw images. This integrated approach establishes a scalable, interpretable solution for automated glaucoma screening and sets the foundation for future multi-modal and clinically deployable diagnostic systems.

Keywords—

Glaucoma detection, fundus imaging, digital image processing, deep learning, EfficientNet, Grad-CAM, CLAHE, optic disc segmentation, medical image analysis.

I. INTRODUCTION

1.1 Background and Motivation

Glaucoma represents a heterogeneous group of optic neuropathies characterized by progressive degeneration of retinal ganglion cells and structural deformation of the optic nerve head (ONH). The disease progresses asymptomatically in early stages, making timely detection crucial for preventing irreversible vision loss. Epidemiological reports consistently identify glaucoma as the second leading cause of blindness worldwide, with increasing prevalence in aging populations and low-screening regions. Clinical assessment typically relies on qualitative and quantitative examination of retinal fundus images to evaluate biomarkers such as the optic disc (OD) shape, optic cup (OC) dimensions, vertical cup-to-disc ratio (vCDR), and retinal nerve fiber layer (RNFL) attenuation. However, conventional assessments remain constrained by inter-observer variability, dependence on ophthalmic expertise, and limited scalability for population-level screening.

1.2 Challenges in Automated Glaucoma Detection

Automating glaucoma detection from fundus images presents two core challenges. First, fundus images exhibit significant heterogeneity across acquisition devices, illumination conditions, patient demographics, and imaging protocols. Public datasets such as AIROGS, REFUGE, ODIR, RIM-ONE, and LAG demonstrate considerable variations in contrast, noise, color distribution, and anatomical visibility, leading to degraded model generalizability Major Datasets Analyzed

Second, glaucoma is inherently multi-stage and structurally progressive, with subtle distinctions between normal, suspicious, early, and advanced stages. Many existing deep learning (DL)-based approaches reduce this to binary classification or depend heavily on pixel-level OD/OC annotations—an expensive and time-intensive requirement. Additionally, the domain shift problem, reported widely across literature, causes a substantial drop in model performance when deployed on previously unseen datasets or real clinical environments

1.3 Limitations of Existing Approaches

Recent advancements such as multi-task segmentation networks, hierarchical DL models, ensemble classifiers, and regression-based vCDR predictors have shown strong performance under controlled dataset conditions. Methods like hybrid clinician-mimicking pipelines, multi-source training, and object-detection-based OD/OC localization significantly improve robustness. However, prior research has identified several persistent limitations:

Annotation bottleneck — segmentation-driven approaches require precise OD/OC masks created by expert ophthalmologists.

Incomplete feature utilization — many models ignore critical glaucoma indicators such as RNFL defects, peripapillary atrophy (PPA), and disc hemorrhages.

Poor handling of edge cases — myopic discs, tilted discs, and atypical ONH morphologies remain underrepresented.

Lack of explainability — black-box classifiers hinder clinical trust and prevent model validation by ophthalmologists.

Weak cross-dataset generalization — performance degradation on external datasets is consistently reported across studies

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These limitations reveal a clear need for a robust pipeline that integrates digital image processing (DIP) techniques with modern deep learning to standardize image quality, mitigate domain shift, and enable clinically meaningful multi-stage classification.

1.4 Purpose of the Present Work

To overcome the aforementioned gaps, this research proposes a comprehensive DIP-driven and explainable deep learning framework for automated glaucoma detection from fundus images. The system incorporates:

Optic disc–centered Region of Interest (ROI) extraction for anatomical precision,

Contrast Limited Adaptive Histogram Equalization (CLAHE) to enhance local contrast in low-quality images,

Non-Local Means (NLM) denoising to suppress sensor- and illumination-induced noise,

A lightweight EfficientNet-B0 classifier fine-tuned for four-stage glaucoma grading, and

Grad-CAM–based explainability to visualize discriminative regions influencing classification. This integrated pipeline aims to improve accuracy, generalizability, and clinical interpretability, providing a technically grounded and deployable solution for real-world glaucoma screening workflows.

The subsequent sections detail the literature landscape, the proposed methodology, and performance evaluation of the system.

2. Literature Review

Author & Year	Dataset(s)	Methodology / Approach	Results	Limitations / Gaps Identified
Sharma et al., 2025	AIROGS, REFUGE, ODIR, Chákṣu	Hybrid multi-model CNN ensemble; OD/OC segmentation + hemorrhage & RNFL detection; Grad-CAM for interpretability	Sensitivity **93.5%** @ 95% specificity	Performance dropped for single CNN; domain shift; requires more diverse datasets
Hemelings et al., 2023	13 data sources: Blue Mountains, Gutenberg, AIROGS, ORIGA, REFUGE, LAG, PAPILA, etc.	Deep regression CNN predicting continuous glaucoma risk (vCDR regression) with standardized preprocessing	AUC **0.976–0.984** across population cohorts	Needs prospective validation; model trained on limited centers; sensitive to external variability
Pascal et al., 2022	REFUGE	Multi-task U-Net for OD/OC segmentation, fovea localization, glaucoma classification	AUC **96.76%**, superior to single-task models	Requires pixel-level annotations; high annotation cost; task interference issues
Xu et al., 2021	Custom 1791 images + 3 external validation sets	Hierarchical interpretable DL model with OD/OC/RNFL segmentation followed by classification	AUC **0.996** (internal), **0.983** (external)	Misses non-structural cues; struggles with low-resolution images; segmentation-dependent
Veturi et al., 2025	27k images + OCT-paired data	OD/OC segmentation + vCDR estimation validated against OCT; trained across 9 public datasets	High agreement with OCT (MAE 0.097, R=0.80)	Limited handling of rare disc shapes; dependent on data diversity; population bias remains

Chaurasia et al., 2024	18,468 images from 20 datasets; external: Drishti-GS	CNN trained on multi-source aggregated dataset	Internal AUROC **0.992**, external AUROC **0.8751**	Significant domain shift; lack of generalization; inconsistent performance on unseen devices
Gao et al., 2024	10 public datasets + REFUGE + UK Biobank	YOLOv7-based OD/OC detection, VCDR computation	Pearson R **0.91**, MAE 0.0347	Does not incorporate RNFL or hemorrhage features; validated mainly on REFUGE-like images
Li et al., 2024 (Review)	50+ datasets (EyePACS, REFUGE, ORIGA, etc.)	Meta-review of preprocessing pipelines (CLAHE, NLM, normalization) and DL backbones	Pooled sensitivity **91.2%**, specificity **89.7%**	Reports major weakness: cross-dataset generalization; need for domain adaptation
Song et al., 2023	ORIGA, DRISHTI-GS, REFUGE	Multi-task CNN with OD/OC segmentation + classification; Grad-CAM	Accuracy **95%**, AUC **0.96**	Limited ethnic diversity in data; lacks robustness across devices
Chen et al., 2024	REFUGE, RIM-ONE, Private dataset	Domain-adaptive EfficientNet; normalization and noise-reduction DIP steps	AUC **0.92**	Domain shift persists; lacks OCT integration for clinical-level sensitivity
Wang et al., 2023	REFUGE, ACRIMA, ODIR	MobileNetV2 with Grad-CAM; CLAHE + OD ROI extraction	Sensitivity **93%**, Specificity **90.8%**	No multi-stage grading; lacks multimodal data fusion
Kumar et al., 2024	ACRIMA, RIGA, Custom staged dataset	Hybrid CNN–Transformer (EfficientNet + ViT) with augmentation	Accuracy **91%**, F1-score **0.89**	Requires stronger interpretability and clinical validation; high compute demand

The recent advancements in automated glaucoma detection illustrate a progressive shift from traditional segmentation-heavy pipelines toward more robust, generalizable, and interpretable deep learning methods. Early segmentation-focused approaches, such as the multi-task U-Net frameworks, emphasized precise extraction of the optic disc (OD) and optic cup (OC), enabling explicit CDR-based diagnosis. Although these methods achieved high accuracy, they relied extensively on pixel-level annotations—making them difficult to scale beyond curated datasets.

To address this limitation, later studies adopted regression-based models that predict continuous glaucoma risk scores directly from fundus images. Notable among these is the G-RISK model, which demonstrated exceptional cross-population generalizability through standardized preprocessing. However, these methods often overlooked structural cues such as RNFL defects and peripapillary atrophy.

Parallel developments explored multi-source training and ensemble learning to combat domain shift—a major bottleneck highlighted across nearly all reviewed research. Multi-dataset CNN models showed significantly improved robustness when tested across heterogeneous populations, though performance still degraded on rare anatomical variations such as tilted or highly myopic discs.

Lightweight architectures such as MobileNetV2 and EfficientNet improved clinical deployment feasibility by offering favorable accuracy–speed trade-offs. Meanwhile, hybrid clinician-mimicking pipelines, hierarchical deep networks, and object detection–based OD/OC localization further enhanced interpretability and diagnostic consistency.

Despite these advancements, the literature consistently identifies five persistent challenges: (1) annotation dependency, (2) insufficient multi-stage glaucoma grading, (3) inadequate cross-device generalization, (4) limited exploitation of subtle clinical biomarkers (RNFL defects, hemorrhages), and (5) lack of transparent decision-making mechanisms. These research gaps motivate the need for an integrated DIP + DL pipeline—like the one proposed in this study—that enhances image quality, performs multi-stage classification, and ensures interpretability using Grad-CAM visualization.

III. PROPOSED METHODOLOGY

The proposed framework integrates Digital Image Processing (DIP) techniques with a deep learning–based multi-stage classifier to achieve robust, explainable glaucoma detection from retinal fundus images. The pipeline consists of four primary components: (1) dataset preparation and labeling, (2) preprocessing and DIP enhancement, (3) deep learning–based four-stage classification, and (4) explainability generation using Grad-CAM.

3.1 Dataset Preparation and Label Mapping

The training dataset consists of publicly available retinal fundus images derived from datasets such as REFUGE, ORIGA, RIM-ONE, and HARVARD, which exhibit considerable variation in color distribution, illumination, resolution, and anatomical clarity. To ensure compatibility across sources, each image is resized to a uniform spatial dimension of 224 × 224 pixels and standardized using ImageNet normalization parameters.

Due to the lack of universal multi-stage glaucoma annotations across datasets, glaucoma severity is mapped into four clinically relevant stages based on documented CDR thresholds and morphological indicators reported in literature:

Normal: vCDR < 0.6

Suspicious: 0.6 ≤ vCDR < 0.7

Early Glaucoma: 0.7 ≤ vCDR < 0.8

Advanced Glaucoma: vCDR ≥ 0.8

Class imbalance inherent in glaucoma datasets is mitigated using class-weighted loss functions and stratified sampling to prevent model bias toward majority classes.

3.2 Digital Image Processing Pipeline

To minimize domain shift and enhance the visibility of subtle pathological features, we design a multi-step DIP pipeline inspired by evidence in prior works highlighting the importance of preprocessing for glaucoma detection

3.2.1 Optic Disc–Centered ROI Extraction

The optic disc is the primary anatomical structure relevant to glaucoma diagnosis. To isolate this region, a disc localization algorithm is implemented using a combination of:

Morphological filtering,

Circular Hough Transform,

Intensity thresholding in the green channel.

The detected disc center is used to crop a circular Region of Interest (ROI), reducing irrelevant background and inter-camera variations. This step improves model focus on clinically meaningful structures and reduces computational load.

3.2.2 Contrast Enhancement using CLAHE

Fundus images often suffer from uneven illumination and low contrast, particularly in community-acquired datasets. Contrast Limited Adaptive Histogram Equalization (CLAHE) is applied to the L-channel of LAB-transformed images to locally enhance contrast without amplifying noise. This results in clearer delineation of the OD/OC boundaries and RNFL patterns.

3.2.3 Noise Reduction via Non-Local Means (NLM) Filtering

To suppress sensor noise, color aberrations, and fine-grain artifacts, Non-Local Means Denoising is applied. NLM preserves edge structures—crucial for accurate CDR estimation—while significantly reducing background noise.

3.2.4 Normalization and Tensor Preparation

Images are normalized to a [0, 1] range and standardized using mean and standard deviation of the ImageNet distribution, ensuring compatibility with the EfficientNet backbone.

This complete DIP pipeline results in a robust, high-quality input tensor, empirically shown to improve the classification performance by ~10–12% in ablation studies from your previous report.

3.3 Data Augmentation Strategy

To address the limited size and heterogeneity of medical imaging datasets, on-the-fly augmentation is performed using Albumentations. The transformations include:

- Random rotation (±15°)
- Horizontal and vertical flips
- Brightness and contrast adjustment (±20%)
- Gaussian noise injection
- Minor affine deformations

These augmentations simulate real-world variability and enhance generalization across unseen populations.

3.4 Model Architecture

The classifier employs EfficientNet-B0, a lightweight yet high-capacity convolutional neural network known for its optimized depth-width-resolution scaling. Key characteristics include:

- Pre-trained ImageNet weights for transfer learning
- A modified classification head comprising Global Average Pooling, Dropout (0.5), and a fully connected layer
- Four output neurons corresponding to the four glaucoma stages
- Softmax activation to predict class probabilities

The model is optimized using:

- AdamW optimizer (learning rate = 1×10^{-3})
- CosineAnnealingLR scheduler for learning rate decay
- Cross-entropy loss with class weights

Training is conducted for 35–50 epochs with early stopping to prevent overfitting.

3.5 Explainability Using Grad-CAM

To ensure interpretability—an essential requirement in clinical decision support—Gradient-weighted Class Activation Mapping (Grad-CAM) is integrated into the inference pipeline. Grad-CAM computes:

1. The gradient of the predicted class score with respect to the last convolutional layer.
2. A weighted activation map highlighting discriminative regions.
3. A heatmap superimposed on the original image.

This visualization enables clinicians to verify that predictions rely on valid anatomical cues such as:

- Optic cup expansion
- Neuroretinal rim thinning
- RNFL loss around the disc

and not on irrelevant background artifacts.

3.6 Deployment Architecture

The complete system is deployed using a FastAPI backend for model inference and a React + TypeScript frontend for user interaction. Both are containerized using Docker for modular, scalable deployment. The API supports:

- Image upload,
- Preprocessing,
- Prediction,
- Grad-CAM heatmap generation.

This architecture supports real-time inference and seamless integration into tele-ophthalmology workflows.

3.7 Summary of Methodological Improvements

The proposed framework offers several innovations over existing work:

- DIP-driven preprocessing pipeline that improves generalization and reduces domain shift.
- Multi-stage classification aligned with clinical glaucoma progression, unlike typical binary systems.
- Explainability integration to enhance trust and diagnostic

clarity.
- Deployment-ready design enabling real-world applicability.
- Together, these components form a cohesive, high-performance, interpretable glaucoma detection system.

4. EXPERIMENTAL RESULTS AND DISCUSSION

4.1 Experimental Setup

Experiments were conducted on a combined dataset consisting of publicly available fundus image sources including REFUGE, ORIGA, and HARVARD. After preprocessing and stratified splitting, the dataset contained approximately 12,000 images, divided into 70% training, 15% validation, and 15% testing. All models were trained on an NVIDIA RTX 3060 GPU (12 GB VRAM) using PyTorch 2.2, with a batch size of 16, learning rate of 1×10^{-3}, and CosineAnnealingLR for dynamic learning-rate decay. The input resolution was fixed at 224 × 224, following the constraints of the EfficientNet-B0 backbone.

The training was performed for 80 epochs, with an early stopping patience of 5 epochs on validation loss. All experiments were repeated three times with different random seeds to ensure stability and reproducibility.

4.2 Quantitative Performance of the Proposed System

The proposed DIP-enhanced EfficientNet-B0 classifier demonstrated strong performance on the four-stage glaucoma classification task. The averaged results from the test set are summarized below, obtained directly from your previous validated results.

Metric	Value
Accuracy	91.7% ± 1.2%
Precision	0.9
Recall (Sensitivity)	0.9
Specificity	0.94
Weighted F1-Score	0.89
AUC (ROC)	0.93

The system effectively differentiated between Normal, Suspicious, Early, and Advanced cases. The confusion matrix revealed that Suspicious vs. Early stages contributed the largest misclassification ratio due to morphological similarities at intermediate CDR values. However, Normal and Advanced categories achieved >95% correct classification, indicating robust detection at the extremes of the disease spectrum.

4.3 Impact of the Digital Image Processing Pipeline

To evaluate the contribution of the proposed DIP pipeline, an ablation study was performed comparing multiple preprocessing configurations:

Model Variant	Accuracy	Δ vs. Baseline
Raw images only	78.40%	–
+ CLAHE + Denoising	85.60%	7.20%
+ ROI Extraction (OD Cropping)	88.90%	10.50%
+ Full Augmentation	91.70%	11.30%

These results clearly demonstrate that CLAHE, Non-Local Means denoising, and optic disc–centered ROI extraction significantly enhance model performance by improving contrast, reducing noise, and standardizing the region relevant to glaucoma diagnosis. The combination of DIP + augmentation yielded the highest accuracy, validating the core motivation behind the proposed pipeline.

4.4 Backbone Comparison: EfficientNet vs. MobileNet

To assess the trade-off between inference speed and accuracy, both EfficientNet-B0 and MobileNet V2 were evaluated using identical data splits and preprocessing:

Backbone	Parameters	Accuracy	Inference Time (ms/image)
MobileNet V2	3.4M	87.20%	22 ms
EfficientNet-B0	5.3M	91.70%	31 ms

EfficientNet-B0 achieved the highest accuracy, making it the most suitable for the final model, while MobileNet V2 provided superior speed for real-time or low-resource deployments. The accuracy-speed trade-off validates the choice of EfficientNet-B0 for the presented multi-stage classification task.

4.5 Explainability Analysis Using Grad-CAM

Grad-CAM heatmaps were generated for all predicted classes to evaluate the interpretability of the system. In correctly classified Early and Advanced glaucoma cases, the activation maps concentrated around:

- Optic cup region
- Neuroretinal rim
- Inferior and superior RNFL sectors

These areas correspond to clinically recognized biomarkers of glaucomatous progression, confirming that the model relies on valid structural cues rather than irrelevant image features.

In contrast, misclassified Suspicious vs. Early samples

showed diffused attention patterns, indicating ambiguous structural presentations rather than model failure. This insight highlights the inherent difficulty of intermediate glaucoma staging and demonstrates the value of explainability tools in validating system decisions.

4.6 Discussion

The experimental results affirm that the integration of DIP techniques with a modern deep learning classifier substantially improves glaucoma detection performance. The system demonstrates:

- High sensitivity and specificity, suitable for screening settings.
- Robust performance uplift after applying ROI extraction and contrast enhancement.
- Effective multi-stage classification, unlike many prior works focused on binary labels.
- Clinically interpretable heatmaps, which are essential for real-world adoption.

However, classification ambiguity remains between intermediate disease stages, consistent with limitations identified in prior literature. Additionally, external validation on geographically diverse datasets is necessary to further evaluate generalizability.

5. CONCLUSION AND FUTURE WORK

5.1 Conclusion

This work presented a comprehensive, Digital Image Processing–enhanced deep learning framework for automated glaucoma detection and multi-stage classification using retinal fundus images. The proposed pipeline integrates three major components: (1) an optics-focused DIP module incorporating CLAHE, Non-Local Means denoising, and optic disc–centered ROI extraction; (2) a fine-tuned EfficientNet-B0 classifier optimized for four-stage glaucoma grading; and (3) Grad-CAM–based explainability to ensure structural and clinical interpretability of predictions.

Experimental evaluation demonstrated that the DIP preprocessing pipeline substantially improves model robustness, yielding an 11.3% increase in accuracy over raw image training and outperforming lightweight models such as MobileNet V2. The system achieved a final accuracy of 89.7%, specificity of 0.94, and AUC of 0.93, confirming the synergy between improved visual quality and deep feature extraction. Furthermore, Grad-CAM visualizations revealed that the model consistently attends to clinically relevant regions such as the optic cup, neuroretinal rim, and RNFL sectors, validating the reliability and transparency of the diagnostic decisions.

Compared to existing literature—which predominantly focuses on binary classification, segmentation-dependent architectures, or regression-based vCDR models—the proposed framework offers an efficient yet interpretable solution capable of performing clinically meaningful multi-stage glaucoma grading. The integration of deployment components (FastAPI backend and React frontend) also positions the system for practical adoption in tele-ophthalmology and large-scale screening environments.

5.2 Future Work

Although the proposed system demonstrates strong performance and interpretability, several opportunities exist for further enhancement and clinical deployment:

5.2.1 Cross-Dataset Domain Adaptation

As highlighted in prior research, domain shift remains a critical challenge in glaucoma detection. Future work will focus on unsupervised or semi-supervised domain adaptation strategies, such as adversarial feature alignment or contrastive learning, to improve generalizability across diverse imaging devices and geographic populations.

5.2.2 Multi-Modal Fusion with OCT and Clinical Metadata

Fundus images alone may not capture early structural changes detectable through Optical Coherence Tomography (OCT). Extending the model to incorporate OCT-derived RNFL thickness, visual field indices, and intraocular pressure (IOP) metadata can significantly enhance diagnostic sensitivity, especially for early and pre-perimetric glaucoma.

5.2.3 Ensemble Modeling for Higher Reliability

While EfficientNet-B0 provides a strong baseline, integrating ensemble approaches—combining CNN, Transformer-based, and regression models—may yield further improvements in robustness, confidence calibration, and multi-stage accuracy.

5.2.4 Lightweight Deployment on Edge Devices

Model compression techniques such as quantization, pruning, or ONNX Runtime optimization will be explored to enable real-time inference on mobile devices, screening kiosks, and portable fundus cameras used in community healthcare settings.

5.2.5 Enhanced Explainability and Clinical Workflow Integration

While Grad-CAM provides saliency visualization, future versions may incorporate more advanced interpretability techniques such as Layer-wise Relevance Propagation (LRP) or vision-language explanations. In addition, integrating patient history logging, result tracking, and clinician feedback loops will improve usability in real clinical workflows.

5.2.6 Large-Scale Clinical Validation

Finally, real-world deployment requires prospective validation through multi-center clinical trials. Target populations should include varying ethnicities, imaging conditions, and ocular comorbidities to rigorously benchmark the model's performance and safety.

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